

# App Dev 2 Project Report

## 1. Student Details

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## 2. Project Details

**Project Title:** Jatayu - Trekking Management Application V2

### Problem Statement:

To build a Trekking Management Application (TMA) web application that allows Admin, Trek Staff, and Users (Trekkers) to interact with the system based on their roles. Also having background jobs (daily reminders, monthly reports, CSV export), API caching, and a Vue.js frontend.

### Approach:

The app was built using Flask REST API with JWT authentication as the backend framework, Vue.js 3 with role-based routing as frontend with Redis for caching and celery for background jobs.

## 3. AI/LLM Declaration

This project was built with AI assistance (Claude) for content, architecture planning, UI/UX design and some functions. All core logic was implemented, integrated, reviewed, tested, and validated by me manually. The extent of AI/LLM usage is around **15-20%**, limited to **code suggestions. documentation formatting** and planning. No use of AI/LLM.

## 4. Technologies and Frameworks Used

| Layer    | Technology      |
|----------|-----------------|
| Backend  | Flask (Python)  |
| Frontend | Vue.js 3 + Vite |

|                 |                                      |
|-----------------|--------------------------------------|
| Database        | SQLite (SQLAlchemy ORM)              |
| Auth            | JWT (Flask-JWT-Extended), role-based |
| Caching         | Redis + Flask-Caching                |
| Background Jobs | Celery + Redis broker + Celery Beat  |
| Styling         | Bootstrap 5 + custom CSS             |

## 5. Database Schema / ER Diagram

Tables:

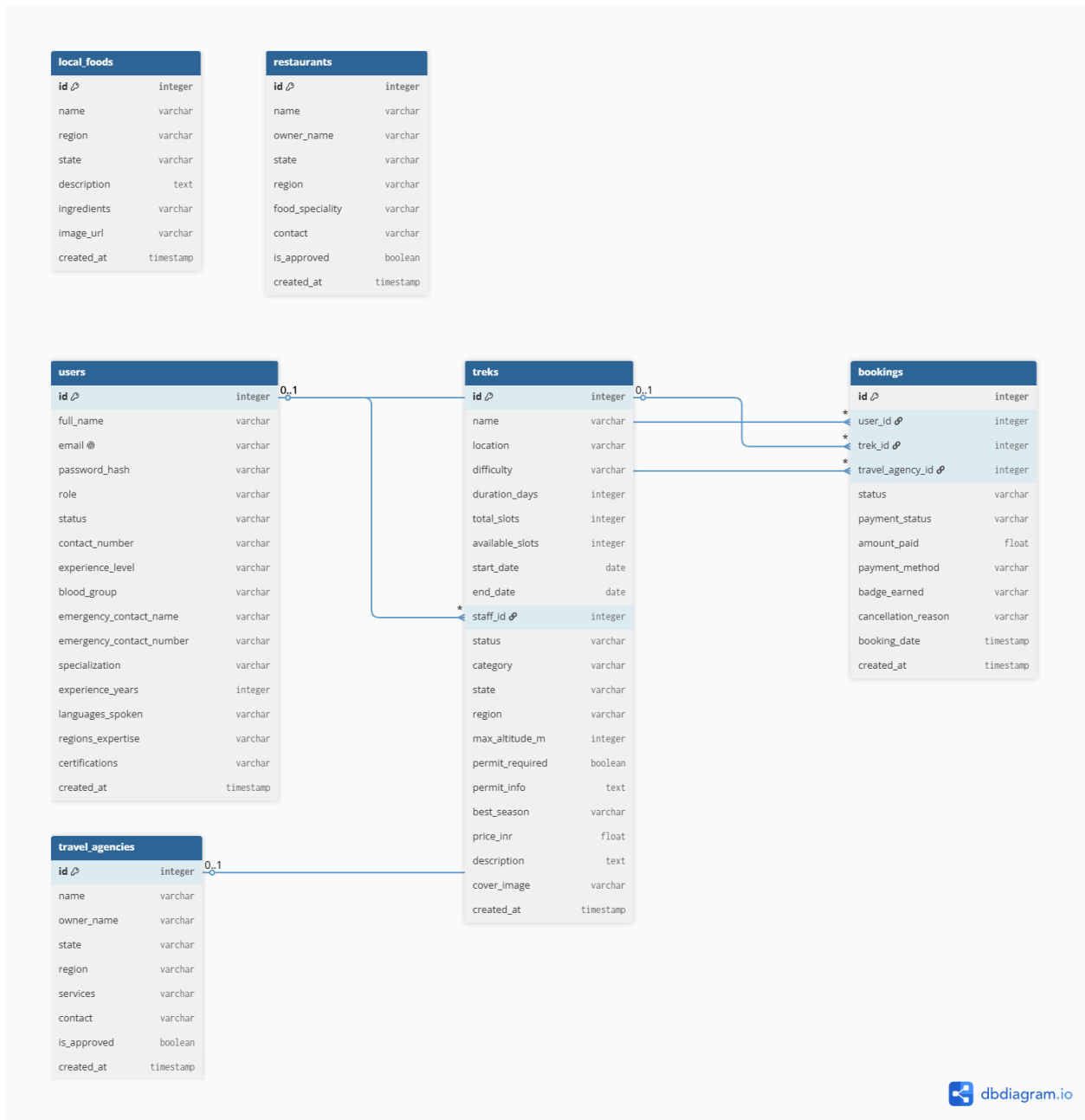
1. users - stores all user accounts (admin, staff, trekker) with role, status, profile details
2. treks - stores trek details (name, location, difficulty, slots, dates, category, cultural info)
3. bookings - logs each trekker's trek booking (status, payment, cancellation reason)
4. restaurants - local eateries near trailheads registered as partners (approval required)
5. travel\_agencies - tour operator companies that handle trek packages (approval required)
6. local\_foods - traditional regional dishes linked to trek regions

Relationships:

- One-to-Many → users → bookings (one trekker can have many bookings)
- One-to-Many → treks → bookings (one trek can have many bookings)
- One-to-Many → users → treks (one staff member can be assigned many treks)
- One-to-Many → travel\_agencies → bookings (one agency can handle many bookings)

## 6. Future Aspects

1. Expand this to every trekking spot in India
2. Include as many trekking partners , restaurant partners, travel agencies, accommodation partners and form a centralised hub system.
3. Provide exclusive experience to the trekkers about Indian tradition but organizing events like pottery , yoga , painting, classical dance , etc.
4. The aim is to provide a holistic joy and extraordinary experience.



## 7. API Resource Endpoints

| Endpoint           | Method | Description             |
|--------------------|--------|-------------------------|
| /api/auth/register | POST   | Register a new trekker  |
| /api/auth/login    | POST   | Login and get JWT token |

|                          |      |                             |
|--------------------------|------|-----------------------------|
| /api/treks/              | GET  | List all open treks         |
| /api/treks/<id>          | GET  | Get full trek detail        |
| /api/admin/dashboard     | GET  | Admin stats and analytics   |
| /api/user/book/<trek_id> | POST | Book a trek                 |
| /api/user/history        | GET  | Get booking history         |
| /api/user/export-history | POST | Export CSV (background job) |

### YAML API Definition File:

Included separately in the submission ZIP as api.yaml.

## 8. Architecture and Features

### Architecture Overview:

- run.py - main Flask application entry point
- /models - database models using SQLAlchemy (User, Trek, Booking, Restaurant, TravelAgency, LocalFood)
- /api - Flask Blueprints for auth, admin, staff, user, treks, restaurants, agencies
- /jobs - Celery background tasks and beat schedule
- /utils - JWT auth decorators and helper functions
- frontend/src/views - Vue.js SPA screens for all three roles

### Implemented Features:

- User registration and login with JWT-based role authentication
- Admin dashboard with trek CRUD, staff management, user management, partner approvals
- Staff dashboard to manage assigned treks, update status, view participants
- Trekker dashboard with trek booking, cancellation, and history tracking
- Browse Treks page with search, filters, map preview, and trek detail view
- Daily reminder emails sent to trekkers 1-3 days before trek start (Celery Beat)
- Monthly admin report with booking stats sent on 1st of every month (Celery Beat)
- User-triggered CSV export of booking history (async Celery task)
- API response caching using Redis to reduce database load

### Additional Features:

- Cultural trek profiles - mythology, local tribes, festivals, best season per trek
- Local food trail - traditional regional dishes linked to each trek's region
- Restaurant and Travel Agency partner registration with admin approval flow
- Responsive dark UI with category-based color themes

## 9. Background Jobs

Three background tasks implemented using Celery + Redis:

| Job            | Trigger                             | What it does                                                                |
|----------------|-------------------------------------|-----------------------------------------------------------------------------|
| Daily Reminder | Every day at 8 AM<br>(Celery Beat)  | Sends email to trekkers whose trek starts in 1, 2, or 3 days                |
| Monthly Report | 1st of every month<br>(Celery Beat) | Sends HTML report to admin with total treks, users, bookings, popular treks |
| CSV Export     | User clicks "Export" button         | Generates booking history CSV and emails it to the trekker                  |

## 10. Frontend Screens

Built with Vue.js 3 + Vite, separate screens for each role:

Public:

- Home Page, All Treks, Trek Detail (with map, gallery, reviews, local food)

Trekker:

- Register / Login, Dashboard, Browse Treks, Booking History, My Profile

Admin:

- Dashboard (stats + charts), Manage Treks, Manage Staff, Manage Users, Partner Approvals, All Bookings

Staff:

- Dashboard, My Treks, Trek Detail (participants + slot management), My Profile

## 11. Video Presentation

**Drive Link:**

 DEMO VIDEO MAD-2 PROJECT.mp4

[https://drive.google.com/file/d/1C0dRL0pb5\\_X3Jz-sA9e3Q42HKEmVb-0F/view?usp=sharing](https://drive.google.com/file/d/1C0dRL0pb5_X3Jz-sA9e3Q42HKEmVb-0F/view?usp=sharing)